

Sri Lanka Institute of Advanced Technological Education

Advanced Technological Institute -Gampaha Project Proposal



Development of a Web-Based Learning Management System (LMS) For Aura Pen Saku Education Institute

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01. Problem Statement

This project is geared towards designing a web-based Learning Management System (LMS) that would help the Aura Pen Tuition Institute solve their inefficient manual processes.

Problem statement: Currently, the Aura Pen Tuition Institute has to keep all records of their students, lesson contents, and quizzes manually since they lack a website and any kind of digitalized processes. Moreover, there is no systematic process in place where the tutes or video lessons can be categorized and shared based on the specific Grade and Subject the students are enrolled in. This process results in chaos and mismanagement of educational data.

Project importance: Addressing the problem is crucial as it will eliminate the inefficiency, mistakes, and wastage of time involved with the manual record-keeping process. It will also secure the educational data and make it easily accessible online to the students who can use their customized dashboard for accessing the relevant lesson content based on their syllabus.

Targeted organization and users: The targeted organization is Aura Pen Tuition Institute. The targeted users are Admin (for system administration), Teacher (to upload grade-wise lesson contents), and Students (who will access their personalized dashboard).

02. Objectives

02.1 General Objectives

The main goal of this project is to develop a comprehensive, web-based Learning Management System (LMS) using Node.js and React.js to automate educational content distribution and student management at Aura Pen Tuition Institute.

02.2 Specific objectives

- ❖ **To design and develop a secure User Authentication system** for students, teachers, and admins to ensure data privacy.
- ❖ **To implement a Grade-wise and Subject-wise Content Management System** that allows teachers to categorize PDF tutes and lesson videos effectively.
- ❖ **To provide a personalized Dashboard** for students to access only the resources relevant to their specific syllabus.
- ❖ **To create an Online Assessment Module** featuring MCQ quizzes to evaluate student progress with automated results.
- ❖ **To establish a centralized MySQL database** for storing and managing institutional data, student records, and resource history accurately.
- ❖ **To improve the efficiency of educational delivery** by replacing the current manual, paper-based method with a modern digital platform.

03. Scope of the project

03.1 In Scope

- **User Authentication:** Implementing a secure way to log into the portal using authentication details for Admin, Teachers, and Students.
- **Dashboard Management:** Implementing a system where the dashboard would dynamically generate depending on the specific grade and subject.
- **Resources Management:** Developing a resource management feature where uploaded PDF tutes and lesson videos are categorized by Grade 10, Grade 11, etc.
- **Online Quizzes:** Implementing an online multiple-choice question quiz where students can attempt quizzes and see instant results.
- **Database Management:** Structuring and managing all database entries and data for the entire institution using a MySQL database.
- **Front-end User Interface:** Building a user interface where the website works effectively on desktops and laptops using React.js.

03.2 Out of Scope

- **Online Payment System:** Any payment or money collection process from parents/students is out of scope.
- **Live Classrooms:** Features such as real-time video conferencing, live streaming, Zoom integration, and similar options would be out of scope.
- **App Development:** There would be no development work done in regard to Android or iOS applications for students/parents/administrators.
- **E-commerce:** Buying or selling anything related to tutes is not within the scope of this project.

04. Proposed Solution

Develop a powerful full-stack application which is suitable for Aura Pen Tuition Institute. This software architecture consists of three-tier: frontend (user-friendly), backend (powerful servers), and database (structured).

Frontend: The UI will be developed using React.js to offer an easy to navigate, faster experience to both students and teachers alike. This part includes a dashboard that will display content according to grades and subjects.

Backend: Server side code will be written in Node.js and Express.js. This layer will manage login functionality, security check for files uploaded by tutes, and calculation of quiz answers correctly.

Database: All institute data will be stored in a centralized MySQL database, which include student records, teacher details, and various learning materials.

Workflow: Instead of recording everything manually, all files will be uploaded on the system itself and managed from there. Then, all students who enroll themselves in the system will have their account created, and the system will automatically present them materials according to the grade they are in. This digital way guarantees 100% accuracy and accessibility to education material at any hour.

05. Key features

- **Feature 1: Login Authentication & Role-Based Authorization**

Login facility will be provided to Admin, Teachers, and Students. According to the respective user role, the user would have permissions (For example, Teachers can upload videos and pdf documents, but students can only read the material allocated by their teacher).

- **Feature 2: Grade-based Dashboard Displaying Videos & Tutorials**

A customized dashboard of each individual student that will display the lesson videos and PDFs according to the grade and subject they belong to.

- **Feature 3: Content Management System (CMS)**

User-friendly CMS interface where teachers or admins can upload, edit, and update all the required course content including external video files from YouTube or Google Drive.

- **Feature 4: Online Quiz System**

Students are provided access to online MCQ quizzes created by their teachers. When a student completes his/her quiz, the score card will appear instantly.

- **Feature 5: Track Student Activities on the Application**

Admin can easily track student activity on the app and keep an eye on which resource is being used and maintain efficient management of student record.

06. Technologies to be Used

To develop this Learning Management System (LMS), the following technologies will be utilized to ensure a high-performance, scalable, and responsive web application.

- **Programming Language:** JavaScript (ES6+).
- **Frontend Technology:** React.js will be used for building a dynamic and responsive user interface. This allows for a smooth user experience on the dashboard.
- **Backend Technology:** Node.js combined with Express.js will be used to handle server-side logic, API requests, and user authentication.
- **Database Management:** MySQL will be the relational database used to store and organize student, teacher, and lesson data securely.
- **Styling & UI:** CSS3 / Bootstrap (to ensure the layout is clean and user-friendly).
- **Version Control:** Git (to manage the project source code).
- **IDE (Integrated Development Environment):** Visual Studio Code (VS Code).
- **Runtime Environment:** Node.js runtime environment.

07. Basic System Requirements

07.1 Hardware Requirement

- ❖ Operating System: Windows 11 Pro
- ❖ Processor: 13th Gen Intel(R) Core(TM) i5-13420H (2.10 GHz)
- ❖ RAM: 16.0 GB
- ❖ System Type: 64-bit operating system, x64-based processor
- ❖ Storage: 512 GB SSD

07.2 Software Requirements

- ❖ Frontend Technology: React.js (Library for building the User Interface)
- ❖ Backend Framework: Node.js with Express.js (For server-side development)
- ❖ Database Management: MySQL (For relational data storage)
- ❖ Runtime Environment: Node.js runtime environment (LTS version)
- ❖ Web Browser: Google Chrome / Microsoft Edge
- ❖ IDE: Visual Studio Code (VS Code)
- ❖ Design Framework: Bootstrap 5 (For styling and responsive layout)

08. Time schedule

	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14	W 15
Select Project title & Proposal Instruction															
Finalize project ideas & initial discussions															
Proposal Submission															
SRS preparation & guidance															
SRS Submission & Review															
Design Diagrams development															
Design Submission															
Start Implementation															
Mid Presentations															
Continue development															
Test Cases Submission & Review															
Report writing guidance															
Project Report Submission															
Presentation preparation															
Final Presentation & Demo															

09. Details of the client

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